SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

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COURSE OUTCOME COVERED

CO1: Design data warehouse architectures with appropriate dimensional models for effective decision support systems. BL1,BL2
Assessment Tool Weightage: 40%

QUESTIONS: 4 **Total Marks:20**

Annexure A
MCQ TEST

Criteria	**Conceptual Understanding(3)**	**Accuracy of Answer(2.5)**	**Application of Knowledge(2)**	**Completeness of Response(1.5)**	**Clarity & Presentation(1)**	***Total(10)***
Weightage	*30%*	25%	20%	15%	10%	*100%*
Q1						
Q2						
Q3						
Q4						

Code of Conduct:

I B.LAKSHMI LOHITHA(192324182)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	**30**	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	**25**	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	**20**	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	**15**	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	**10**	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

ETL (Extract, Transform, Load)

Name : B. Lakshmi Lohitha
Reg No: 192324182

Introduction

What is ETL?

ETL (Extract, Transform, Load) is a process used to collect, clean, and store data from different sources into a Data Warehouse. It helps organizations convert raw data into meaningful information for reporting and decision-making.

Why is ETL Important?

- Collects data from multiple sources. w
- Cleans and transforms data into a standard format.
- Stores reliable data in a Data Warehouse.
- Supports Business Intelligence (BI) and data analysis.
- Enables faster and more accurate business decisions.

How ETL Works?

ETL works in three simple steps:

- **Extract**

Collect data from different sources.

Example: Sales, Customer, and Payment databases.

- **Transform**

Clean the data.

Remove duplicate records.

Convert data into a common format.

- **Load**

Store the processed data in the **Data Warehouse**.

ETL Tools and Technologies

- **Apache NiFi**

  Used for real-time data flow management and data integration.
- **Informatica PowerCenter**

  A popular ETL tool for enterprise data integration and transformation.
- **Talend**

  Open-source ETL tool used for data processing and migration.
- **Microsoft SSIS**

  Helps extract, transform, and load data in Microsoft environments.
- **Apache Spark**

  Processes large-scale data efficiently for big data analytics.

Advantages of ETL

- Improves data quality by removing errors and duplicates.
- Saves time in data processing.
- Provides accurate and reliable reports.
- Supports better business decisions.
- Organizes data in a standard format.
- Enables faster data analysis.
- Increases business efficiency and productivity.

Real-Time Applications of ETL

- **E-Commerce :**

  ETL analyzes customer searches, purchases, and reviews to provide recommendations and improve sales.

- **Banking:**

  ETL processes transaction data to detect fraud, manage risks, and improve financial services.

- **Healthcare**

  ETL integrates patient records and medical data to support diagnosis and better healthcare decisions.

- **Transportation**

  ETL processes GPS and traffic data to provide real-time navigation and route optimization.

Future Scope of ETL

The future of ETL focuses on faster, smarter, and automated data processing. With the growth of big data, cloud computing, and artificial intelligence, ETL systems will be able to handle large volumes of data in real time. AI-based ETL will improve data cleaning and transformation, while cloud-based solutions will provide scalable and flexible data integration for modern businesses.

Thank you…